



Module 12 Assignment

Individual Project Proposal

ALY 6080: Integrated Experiential Learning

Assignment Completion Date: 12/10/2022

Instructor: Professor Jayash Koshal

Term: Fall 2022

Author:

Brian Griffin

I. Statement of Purpose

Within the financial services industry, registered representatives sell a variety of products to individual investors. These products include stocks, bonds, mutual funds, ETFs, bank products, insurance products, options, and annuities, among many others. The purpose of purchasing these products varies among investors, however retirement savings, college savings, and philanthropic causes are some of the most common driving factors for making these investments. Registered representatives assist investors in making these investments based upon multiple factors, although the two most important factors are risk and return. According to the Corporate Finance Institute, risk and return are highly correlated. Typically, the more return an investor wishes to pursue, the more risk they must assume in their portfolios (Corporate Finance Institute, 2022). In achieving returns, registered representatives and their clients must decide whether they will pursue an active or passive investment management strategy. Active management is an approach where an investor intentionally selects the investments that make up their portfolio, through security selection or asset allocation. Sustainability analysis is also another factor that many investors consider whom utilize the active management approach. Conversely, passive management is a strategy in which the investor tracks a market-weighted index, portfolio, or benchmark, often trying to replicate the performance of the broader market (Investment Advisor Association, 2022). The purpose of my proposal is to suggest the development of an analytic tool that will assist in predicting whether an investor's active management strategy will outperform passive management based on selected parameters over a long-term approach. This will assist the investor and their registered representative to select the best approach implementing the most effective investment strategy.

II. Project Scope

The scope of this project will be limited to tracking and attempting to predict the performance of mutual funds and ETFs versus market benchmarks. An ETF (exchange traded fund) is similar to a mutual fund in that it pools multiple securities together in one instrument, but differs in that it is traded on a stock exchange (Vanguard, n.d.). The scope of the project is being limited to these instruments since it would be too large of a project to address additional investment vehicles in one effort given resources and time constraints. The goal of the project is to determine if an investor should choose a specific actively managed mutual fund / ETF versus the corresponding passively managed product covering the same benchmark. The primary deliverable that will be accomplished is a model, produced in the form of a Python script, that will make a prediction if a single actively managed mutual fund (or ETF) will outperform a single passively managed mutual fund (or ETF) on a risk-adjusted basis over a specified period - of-time.

III. Background Research

It would be a good idea to implement this project because it will allow registered representatives to gain insights into what investments they should make for their clients based on the results derived from the model being proposed. There is no crystal ball in the world of investments, but the proposed model will assist the registered representative in making a data-driven, informed decision, in the best interests of their client. I have performed research in the form of annotated bibliographies alongside my XN Project Teammates, along with additional individual research, which will be summarized below.

Annotated Bibliography I: *The Hidden Costs of Maintaining Registered Rep Licenses*

In this article, Alan Foxman offers his thoughts and ideas on the current value of registered representatives and their position in the financial services industry. From his perspective, registered representatives are a part of an evolving landscape which only the highest performers have a secured position within a firm. Additionally, Foxman explains that registered reps are likely to move around and work for different companies throughout their career as a licensed professional (Foxman, A., 2021). In relation to my proposal, I believe that an analytic tool developed to help predict whether an active strategy would outperform a passive strategy would directly benefit registered representatives in assisting them to make decisions for their clients, helping the reps become high performers, allowing them to succeed in the industry.

Annotated Bibliography II: *Investment advisor industry grows for 9th consecutive year*

In this article, Michael S. Fisher explains that the number of investment advisors registered with the SEC, along with the assets they and their firms managed, increased to record highs during 2020. Along with this growth, the industry has seen trends towards representative compensation shifting more towards performance fees and fixed fees, while commissions have become less common. The industry's growth shows that investors and investment advisory firms recognize the value of fiduciary advice (Fischer, M. S., 2021). In relation to my proposal, I believe that the growth in registered representatives, along with compensation shifting towards performance-based fees, makes this market a prime target market for the analytic tool I am proposing on delivering. The analytic tool can be marketed towards a growing registered rep industry, where outperforming other representatives, with the help of this tool, is rewarded with higher performance-based fees.

Additional Research: *Active vs. Passive Investing: How to Choose an Investment Strategy for You*

According to insights provided by Marcus, the online-only banking arm of Goldman Sachs, there are benefits and disadvantages of both active and passive investment strategies. Passive management has benefits of lower fees, tax efficiency, and transparency (being able to identify exactly what securities are in your portfolio). Disadvantages include less control over the portfolio, and smaller short-term returns. Conversely, active investing provides more flexibility, and risk management options. Disadvantages to active management include higher fees, and ambiguity on future performance of the fund (Marcus by Goldman Sachs, 2020). In relation to my proposal, an analytic tool that can assist in predicting performance of an actively management fund versus a passively managed fund, can help provide insights to a registered representative or investor in determining what option might be preferable in a specific situation, active or passive, in relation to the other benefits and disadvantages of each strategy that must be considered in the decision.

IV. Design and Data Collection Methods

Data that can be utilized for this proposal is publicly available across both actively managed mutual funds, ETFs, and benchmarks. The data points that will need to be collected include historical performance returns of the funds and benchmarks (net of fees), along with a risk measurement of the funds in the form of a standard deviation data point. Publicly available mutual fund data can be sourced from a variety of sources, including the SEC, the Investment Company Institute, and companies that aggregate this data (often for a fee) such as Factiva, Morningstar, Standard & Poor's, and Thomson One, among many others (Tulane Library Guides, n.d.). I propose Morningstar as a primary source to be used for this data. Morningstar

provides API's that provides thousands of mutual fund data points along with benchmark and index data points that could be consumed in the model being proposed to perform analysis (Morningstar, n.d.). Additionally, interest rate data will be collected from the Federal Reserve Bank. Interest rates have a significant impact on investment performance based on the securities comprised within a mutual fund.

V. Methodology, Methods, and Strategies

A predictive model is being proposed as the deliverable for this project. The predictive model will be delivered in the form of a Python script. A Python script will allow the analysis to be performed on multiple fund comparisons and the code can be reused over time. The model will analyze the relationship between mutual fund performance returns (dependent variable) among multiple independent variables, using multiple linear regression. Multiple linear regression is used to predict the outcome of a variable based on two or more independent variables (Dawson, C., 2021). The independent variables being utilized in the model include: standard deviation data points (historical, over time) of the mutual fund being analyzed, index sector returns defined by "Select Sector SPDR ETF's" (historical, over time), and interest rates defined by Federal Reserve Federal Funds Rate (historical, over time). Select Sector SPDR ETF returns are unique exchange traded funds that divide the S&P 500 into eleven index funds based on industry (Sector SPDR, n.d.). The Federal Funds rate is the target interest rate set by the Federal Open Market Committee (FOMC), which is the rate which commercial banks borrow and lend their excess reserves to each other overnight (Chen, J., 2022). The federal funds rate impacts broader financial conditions such as growth, inflation, and therefore the stock market. Within the Python script, multiple libraries will be utilized. Two of the primary libraries will be Pandas, for handling the dataset, and scikit-learn, for performing the multiple regression. Within

scikit-learn, the LinearRegression module will be utilized. The model will first be performed for a mutual fund with an active strategy. The same model can be used for an index (passive strategy) corresponding to the active strategy being analyzed. The results can be compared, assuming the data inputs cover the same time periods.

Model Specifications

Model: Multiple Linear Regression

Scripting Language: Python

Package: pandas, sklearn (modules: model_selection, LinearRegression)

Dependent Variable: Mutual Fund Performance Return

Independent Variables: standard deviation of fund (or index), index sector returns (11 returns, each sector), Federal Funds Rate

VI. Definition & Measurement of Success

To ensure that my proposal meets the sponsor's needs, I will need to define and measure success so that I will know what needs to be achieved upon completion of the project and that the model meets requirements. First, I think that the performance of my model over a specific time-period will need to be established. For example, what is the performance of the fund analyzed, benchmark analyzed, over a 5-year period? I think 5-years is the most practical period to measure performance because it avoids short-term strategy making it sustainable, but not so long-term that the investor must wait to realize returns. Additionally, I think success will be measured in whether the correct investment was chosen, either the mutual fund or the benchmark. If the mutual fund is chosen, it must return alpha, or a return above the benchmark. If the benchmark is chosen, it must outperform the mutual fund. The purpose of this analytic model is to provide a return for the investor. Therefore, evaluation criteria can be summarized as

follows: was the correct investment chosen (mutual fund or benchmark) over a 5-year time-period based on whether the investment chosen outperformed its alternative based on performance returns over that period. Clients of the analytic tool being proposed will need to be informed that 5 years' worth of performance will need to be collected before making an informed assessment if evaluation criteria has been met. This ensures a long-term investing approach, which is the goal of this analytic tool.

VII. Conclusion

The goal of this proposal is to produce an analytical model that will determine if an active strategy is performed over a passive strategy based on selected parameters. The model will predict mutual fund performance (active strategy) or index performance (passive strategy) by utilizing a multiple linear regression model based on standard deviation, index sector returns, and the federal funds rate. The model will be flexible in that it can analyze results for any mutual fund that has publicly available historical results. The results of this model will assist registered representatives and their clients in making informed, data-driven investment decisions, to reach optimal returns.

Reference:

About select sector SPDR ETF funds. About Select Sector SPDRs. (n.d.). Retrieved October 23, 2022, from <https://www.sectorspdr.com/sectorspdr/features/about>

Chen, J. (2022, September 22). *Federal funds rate: What it is, how it's determined, and why it's important.* Investopedia. Retrieved October 23, 2022, from <https://www.investopedia.com/terms/f/federalfundsrate.asp#:~:text=The%20federal%20fu,nds%20rate%20is%20the%20interest%20rate%20that%20banks,at%20a%20Federal%20Reserve%20Bank.>

Dawson, C. (2021, February 1). *Understanding multiple linear regression.* Medium. Retrieved October 23, 2022, from <https://medium.com/swlh/understanding-multiple-linear-regression-e0a93327e960>

ETFs - exchange traded funds - what are etfs? Vanguard. (n.d.). Retrieved October 23, 2022, from <https://investor.vanguard.com/investment-products/etfs>

Fischer, M. S. (2021, June 28). *Investment advisor industry grows for 9th consecutive year .* ThinkAdvisor. Retrieved October 6, 2022, from <https://www.thinkadvisor.com/2021/06/28/investment-advisor-industry-grows-for-9th-consecutive-year/>

Foxman, A. (2021, June 14). *The Hidden Costs of Maintaining Registered Rep Licenses.* Retrieved October 2, 2022, from <https://www.advisorperspectives.com/articles/2021/06/14/the-hidden-costs-of-maintaining-registered-rep-licenses>

Investment Data: Mutual Fund databases. Morningstar. (n.d.). Retrieved October 23, 2022, from <https://www.morningstar.com/products/data>

Marcus by Goldman Sachs. (2020, September 13). *Active vs. passive investing: How to choose an investment strategy for you.* Marcus by Goldman Sachs®. Retrieved October 23, 2022, from <https://www.marcus.com/us/en/resources/investing/active-vs-passive-investing>

Mutual funds: Sources for Mutual Funds Data & Information. Library Guides. (n.d.). Retrieved October 23, 2022, from <https://libguides.tulane.edu/mutualfunds>

Risk and return. Corporate Finance Institute. (2022, February 8). Retrieved October 23, 2022, from <https://corporatefinanceinstitute.com/resources/knowledge/trading-investing/risk-and-return/>

What is active management? - investment adviser association. Investment Adviser Association -. (2022, April 8). Retrieved October 23, 2022, from <https://investmentadviser.org/active-managers-council/what-is-active-management>